Journal of Theoretical Biology 455 (2018) 7–15

Contents lists available at ScienceDirect

Journal of Theoretical Biology

journal homepage: www.elsevier.com/locate/jtb

A solution for private assessment in indirect reciprocity using solitary observation

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ARTICLE INFO

Article history:
Received 7 April 2018
Accepted 19 June 2018
Available online 3 July 2018

Keywords:
Evolution of cooperation
Indirect reciprocity
Private information
Evolutionary game theory

ABSTRACT

Although indirect reciprocity is a fundamental mechanism in the evolution of human cooperation, most studies assume public assessment in which individuals are not permitted to obtain private assessments of others. Existing studies on private assessment have used individual-based simulations because of the analytical difficulty involved. Here, we develop an analytical method using solitary observation to solve private assessment in indirect reciprocity problem without any approximation. In this study, we formulate a model of solitary observation and calculate the replicator dynamics systems of five leading norms of indirect reciprocity. Indirect reciprocity in private assessment provides a different result to that in public assessment. According to the existence proofs of cooperative evolutionarily stable (CES) points in the system, strict norms (stern judging and shunning) have no CES point in private assessment, while they do in public assessment. Image scoring does not change the system regardless of the assessment types because it does not use second-order information. In tolerant norms (simple standing and staying), the CES points move to co-existence of norms and unconditional cooperators. Despite the fact that there is no central coercive assessment system in private assessment, the average cooperation rate at the CES points in private assessment is greater than that in public assessment. This is because private assessment gives unconditional cooperators a role. Our results also show the superiority of the staying norm. Compared with simple standing, staying has three advantages in private assessment: a higher cooperation rate, easiness of invasion into defectors, and robustness to maintain cooperative evolutionarily stable situations. Our results are applicable to general social dilemmas in relation to private information. Under some dilemmas, norms or assessment rules should be carefully chosen to enable cooperation to evolve.

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1. Introduction

Indirect reciprocity (Ghang and Nowak, 2015; Leimar and Hammerstein, 2001; Milinski et al., 2002; Nowak and Sigmund, 1998a; 1998b; 2005; Ohtsuki et al., 2009; Rockenbach and Milinski, 2006; Sasaki et al., 2016; Suzuki and Kimura, 2013) is a fundamental mechanism in the evolution of cooperation even in modern society which has highly mobile and large-scale relationships among individuals. Assessments of unrelated individuals are an essential mechanism for the avoidance of free riders, and thus it is considered that indirect reciprocity requires a shared reputation system that distributes images of all individuals in a society (Grimalda et al., 2016; McNamara and Doodson, 2015; Sommerfeld et al., 2007; Swakman et al., 2016). Therefore, the majority of theoretical studies on indirect reciprocity have assumed public assessment (Brandt and Sigmund, 2004; 2006; Chalub et al., 2006; Fishman, 2003; Leimar and Hammerstein, 2001; Masuda and Ohtsuki, 2007; Ohtsuki and Iwasa, 2004; 2006; 2007; Panchanathan, 2011; Panchanathan and Boyd, 2003; 2004; Sigmund, 2010; Watanabe et al., 2014). However, this is an over-simplification because an image of an individual is not necessarily the same for all individuals. Several studies have focused on private assessment (Martinez-Vaquero and Cuesta, 2013; Ohtsuki et al., 2015; Olejarz et al., 2015; Sigmund, 2012; Uchida, 2010; Uchida and Sasaki, 2013; Uchida and Sigmund, 2010), in which they relax such an unnatural assumption. Unfortunately, solving the problem of private assessment in indirect reciprocity is difficult because the number of equations goes

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https://doi.org/10.1016/j.jtbi.2018.06.018
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to infinity. Thus, to date, studies on private assessment have used either numerical analysis or individual-based simulations (Okada et al., 2017; Yamamoto et al., 2017).

Previous studies on private assessment in indirect reciprocity using individual-based simulations have produced results that differ significantly from those related to public assessment. Okada et al. (2017) reveal three features: narrow and unstable cooperation, stable coexistence of discriminators and unconditional cooperators, and Pareto improvement. Although their pioneering work should be acknowledged, rigorous analysis using a theoretical solution is needed for general situations.

Here, we develop an analytical method using solitary observation to solve the problem of private assessment in indirect reciprocity. Any model of private assessment without restricted conditions cannot be solved analytically because this would require the solving of an infinite number of equations. This is because the definition of the conjunctive probability of ν players whose images of a specified player are the same needs the conjunctive probability of $\nu+1$ players when the number of game observers is infinite. The extreme case of this restriction is a solitary observation in which the number of observers per game is set at one. Although this might be an extreme assumption, a theoretical comparison between public and private assessment is important for rigorous discussion.

In this study, we consider the five leading norms in indirect reciprocity studies: image scoring (Nowak and Sigmund, 1998b; Wedekind and Milinski, 2000), simple standing (Milinski et al., 2001; Sugden, 1986), stern judging (Kandori, 1992; Pacheco et al., 2006; Santos et al., 2016), shunning (Takahashi and Mashima, 2006), and staying (Nakai and Muto, 2005; 2008; Okada et al., 2017; Sasaki et al., 2017). We formulate a model of solitary observation and calculate the replicator dynamics systems of the five leading norms of indirect reciprocity. The analysis of a solitary observation shows that cooperative evolutionary stable strategies depend on the cost benefit ratio of a social dilemma game in private assessment, while this is not the case in public assessment. Our results confirm the insights of Okada et al. (2017).

2. Results

2.1. Giving games and solitary observation

We consider players who play giving (donation) games. A donor playing a game decides whether to contribute (C) or do nothing (N). If the donor chooses C, she or he must pay a fixed amount to a recipient. The recipient receives a benefit if and only if the donor contributes. We set the cost benefit ratio of a game to $r>1$. If the donor chooses N, nothing happens. Self-interested myopic players choose N because contributors do not benefit from their own contribution, and thus our model reveals a social dilemma. Players in the model consist of three norm-adopters: unconditional cooperators (X) who always contribute, unconditional defectors (Y) who never contribute, and discriminators denoted as Z. The discriminators have their private binary images (good or bad, denoted as G and B, respectively) of all the other players and only contribute to those whose images are good, thus they are called conditional cooperators.

All of the players face two types of errors: those in implementation and those in assessment. The donor defects in contradiction to her or his intention to contribute with a probability, e_1, and the observer oppositely mistakes the assessment for the donor with a probability, e_2. Let $\bar{e_1}=1-e_1$, $\bar{e_2}=1-e_2$, and $\epsilon=\bar{e_1}\bar{e_2}+e_1e_2$. For simplicity, we assume that $0<e<1/2<\epsilon<1$ is strictly satisfied where e is replaced by e_2.

In solitary observation, a game can only be observed by one observer. In each game, a donor (D), a recipient (R), and an observer (V) are randomly chosen. A player may fill one, two, or all three of those roles while the observer must be a Z player. In the game, only the observer is given a chance to update the donor's image, while the other discriminators have no such opportunity. Let $I_{jk}\in\{G, B\}$ be player k's image as assessed by player j. For example, I_{DR} is a recipient's image assessed by a donor in a game.

Table 1
Norms considered in this study.

Assessment rules	Donor's action (C/N) and Recipient's image (G/B)			
	C/G	N/G	C/B	N/B
Image scoring	G	B	G	B
Simple standing	G	B	G	G
Stern judging	G	B	B	G
Shunning	G	B	B	B
Staying	G	B	K	K

Note: "K" means that the discriminator does not update the donor's image regardless of one's play.

2.2. Replicator dynamics systems in the games

An infinite number of well-mixed players play games repeatedly using a continuous-entry model (Brandt and Sigmund, 2005). Let x, y and z be the fractions (population ratios) of X, Y, and Z, respectively, where $x+y+z=1$ is always satisfied. To explore the evolutionary dynamics of the private images, we consider an analysis of the marginal value of a good reputation (Ohtsuki et al., 2015), where no two players can ever meet more than once due to the assumption of an infinitely large population, and thus the chance of direct reciprocity is excluded from the model. In this framework, the time scale for natural selection is much slower than that of social interactions and image updates. Therefore, we can always assume that the frequency of good players is at its equilibrium value, that is, the expected probability that a players image is good has converged to a steady state. Then, we can define the parameters regarding the frequencies of good players in the steady state. Let g be the fraction of good players assessed by Z. This fraction is decomposed into g_x, g_y and g_z, where g_s is the fraction of good players with strategy s in the set $S=\{X,Y,Z\}$. $g=xg_x+yg_y+zg_z$ is always satisfied.

We consider a replicator equation system (Hofbauer and Sigmund, 1998). Let P_s be the expected payoff per the cost of the game of an s strategist where $s\in S$. The replicator dynamics are described as $\dot{x}=x(P_X-\bar{P}), \dot{y}=y(P_Y-\bar{P})$, and $\dot{z}=z(P_Z-\bar{P})$, where $\bar{P}=xP_X+yP_Y+zP_Z$ is the average payoff over the population. The expected payoffs of the three strategists are:

$$P_X = r(x+zg_x)-1$$
$$P_Y = r(x+zg_y)$$
$$P_Z = r(x+zg_z)-g.$$

where we omit the factor $\bar{e_1}$.

2.3. The five leading norms

In this study, we consider the five norms shown in Table 1. These norms have a common feature in terms of their assessment rules: contribution to a good recipient is assessed as good and defection to a good recipient is assessed as bad. The "image-scoring" norm is that cooperation is assessed as good, while defection is assessed as bad regardless of the recipient's image. The most tolerant norm, which is called "simple standing," is that both cooperation and defection to a bad recipient are assessed as good, while the stricter norm, which is called "stern judging," is that contribution to a bad recipient is assessed as bad while defection is assessed as good. The strictest norm is called "shunning," where any action to

Table 2
Features of cooperative evolutionarily stable points for the five norms. We refer to the case of public assessment in Sasaki et al. (2017). Note that when $e_1 = e_2 = 0$, the conditions of existing CES points are $r > 1$, $r > 1$ and $r > \infty$, in the staying, simple standing and shunning norms, respectively.

	Condition of existing cooperative evolutionarily stable point	The average non-cooperative rate on the point if existing	Same rate in the case of public assessment
Staying	$r > \frac{1}{\epsilon - e}$	$\frac{r\epsilon - 1 + (1-\epsilon)(\epsilon - e)}{(r-1)(\epsilon - e)(r\epsilon - 1 + \epsilon)} \times \frac{r(1-\epsilon)}{r-1}$	$1 - \epsilon$
Simple-standing	$r > \frac{1}{\epsilon - e}\left(1 + \sqrt{\frac{1-\epsilon}{1-e}}\right)$	$\frac{1 - \epsilon + \frac{r(2-\epsilon-e)-1}{\epsilon - e}}{1 - \epsilon + r^2(1-e) - r} \times \frac{r(1-\epsilon)}{r(2-\epsilon-e)-1}$	$\frac{1-\epsilon}{2-\epsilon-e}$
Stern-judging	(non-existence)	(non-existence)	$\frac{1-\epsilon}{2-\epsilon-e}$
Shunning	$r > \frac{1 + \sqrt{1 - 4e(\epsilon - e)}}{2e(\epsilon - e)}$	$\frac{2(\epsilon - e) - 1 + \sqrt{1 - 4e(\epsilon - e)}}{2(\epsilon - e)}$	$\frac{1-\epsilon}{1-\epsilon+e}$
Image-scoring	(non-existence)	(non-existence)	(non-existence)

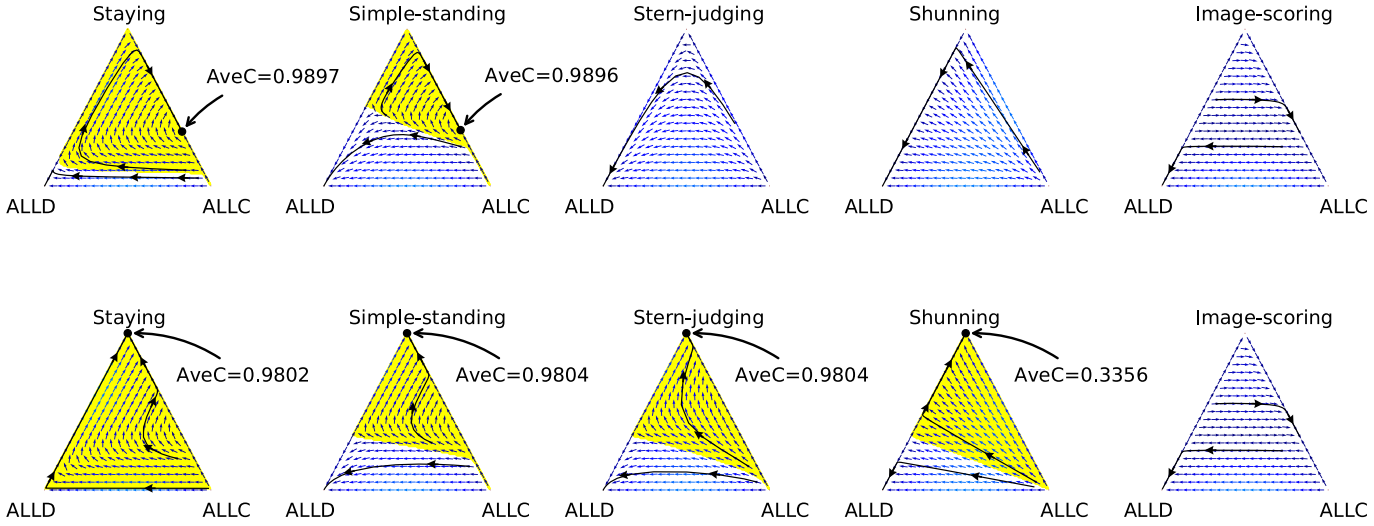


Fig. 1. Replicator dynamics of the five norms in private and public assessments. The upper panels show the results in the case of private assessment and the lower panels show those in the case of public assessment referred in Sasaki et al. (2017). The triangles describe a simplex of the state space, $\{(x, y, z) | x + y + z = 1\}$, where x, y, and z are non-negative real numbers denoting the frequencies of unconditional cooperators (ALLC), unconditional defectors (ALLD), and norm adopters, respectively. An arrow in the triangles shows the direction of replicator dynamics at each point. Some trajectories following the dynamics are also drawn. If a CES point exists, the average rate of cooperation (AveC) at that point is calculated and the basin of attraction is shown in yellow. The area ratios of the basin are 84.81% and 45.46% in the staying and simple-standing norms, respectively, in the case of private assessment, while they are 99%, 58.07%, 59.23%, and 65% in the staying, simple-standing, stern-judging, and shunning norms, respectively, in the case of public assessment. The parameter values are $r = 3$, $e_1 = 0.01$, and $e_2 = 0.01$.

a bad recipient is assessed as bad. In the "staying" norm, the image of the donor remains unchanged if the potential recipient has a bad image.

2.4. Analytical solutions of the systems

We formulate a model of solitary observation and solve its equation systems for the five norms in Materials and methods. First, similar to cases of public assessment presented by Sasaki et al. (2017), a population consisting of 100% defectors ($y = 1$) is an asymptotically stable fixed point for any norm. This is almost trivial, so we then explore the other asymptotically stable fixed points. We call them cooperative evolutionarily stable (CES) points. In staying, there is a CES point on the line $y = 0$ if r exceeds a threshold. The point consists of unconditional cooperators and staying-norm adopters. The case of simple standing also has a CES point on the line $y = 0$ if r exceeds a threshold. In stern-judging, there is no CES point in private assessment regardless of r while there is a CES point in public assessment. In shunning, there is a CES point in private assessment if r is extremely large. In image scoring, the dynamics in the case of private assessment is the same as that in the case of public assessment. Table 2 summarizes the features of the CES points for the five norms.

Fig. 1 shows the replicator dynamics systems of the five norms in both private and public assessments for a specific parameter setting. Comparing public and private assessments, our results confirm Okada et al.'s (Okada et al., 2017) three insights. First, the cooperative stable norms are more restricted in private assessment than those in public assessment. As Fig. 1 shows, stern judging and shunning have no CES points in private assessment, while they do have CES points in public assessment. Second, in private assessment, the cooperative stable norms (simple standing and staying) are more likely to coexist with unconditional cooperators in contrast to the public assessment case. Third, the average cooperation rate at the CES points in private assessment is higher than that in public assessment.

Moreover, our rigorous analysis reveals that, in staying and simple standing, the average cooperative rate at the CES point depends on the cost-benefit ratio of the game (r) in private assessment, while it remains constant regardless of r in public assessment. Fig. 2 shows the results when the scopes of e_1 and e_2 are 0.1% to 5%. Note that in Fig. 2, the average non-cooperative rates are compared instead of the average payoffs. This is because the average payoff U is a negative linear function of the average non-cooperative rate k, as $U = (r - 1)(1 - k)$. As can be seen from Fig. 2, if r is less than about 2, public assessment provides bet-

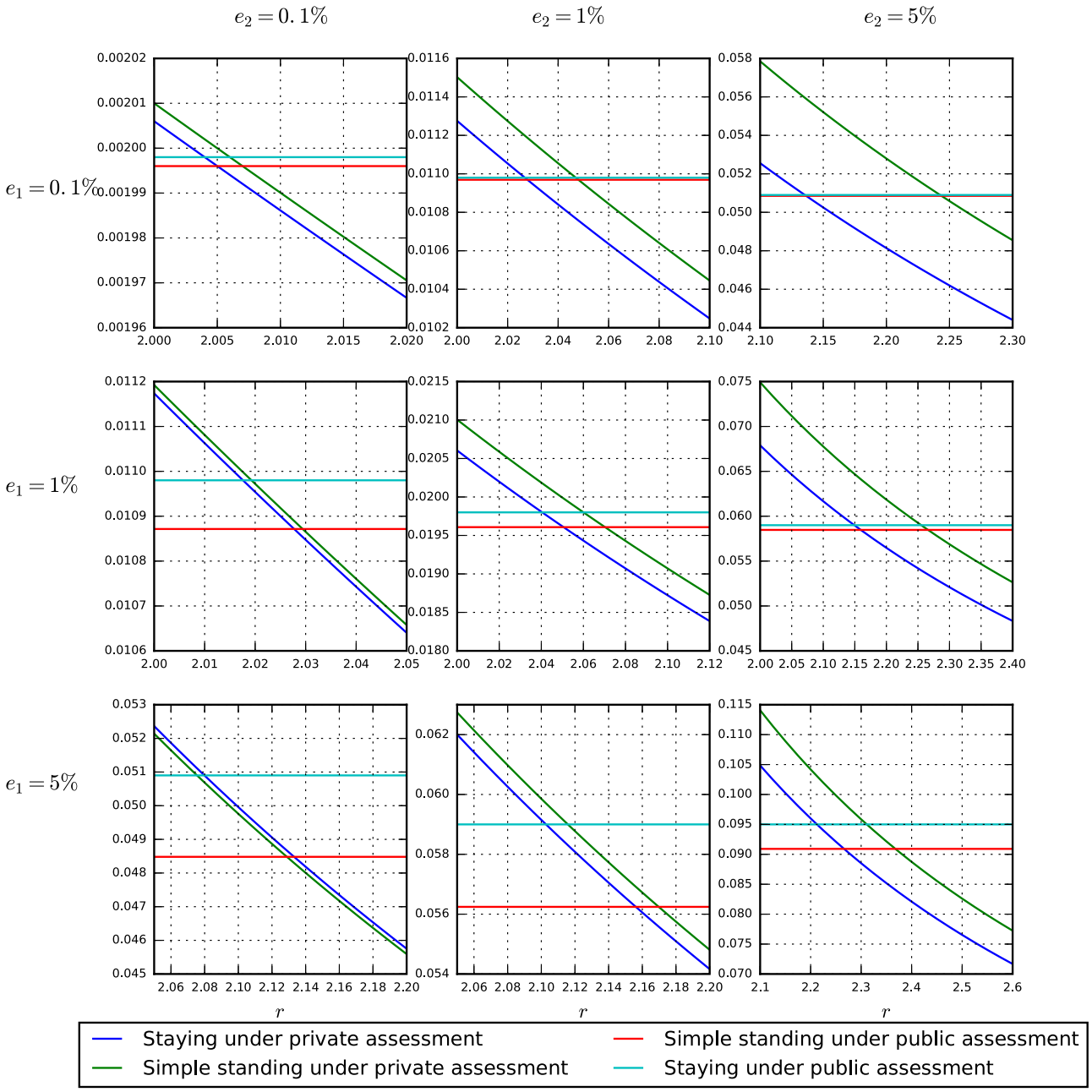


Fig. 2. The average non-cooperative rates at the cooperative evolutionarily stable point. Each panel shows one of the combinations of $e_1 \in \{0.1\%, 1\%, 5\%\}$ and $e_2 \in \{0.1\%, 1\%, 5\%\}$. The norms in public assessment do not depend on the cost benefit ratio of the game (r), whereas they do in private assessment.

ter performance than private assessment, while if r is greater than about 2.5 the reverse applies.

2.5. Superiority of staying

Our theoretical analysis also reveals that staying has some superiority in relation to the evolution of cooperation. First, staying has the widest area ratio for the basin of attraction of all five norms in both public and private assessments, as shown in Fig. 1. In particular, the distance from the point of 100% defectors ($y = 1$) to the basin of attraction is the shortest. This suggests that staying is the easiest norm to be able to invade into a population of 100% perfect defectors. Second, in simple standing, the CES point is located almost at the border of the basin of attraction, while that is not the case in staying. This suggests that staying is more robust over a turbulence by mutations than simple standing at the CES points. Third, at the CES point, the population in staying provides better cooperation than that in simple standing except for $(e_1, e_2) = (5\%, 0.1\%)$ in Fig. 2. This is in contrast to the case of public assessment because a population in which everyone adopts the staying norm does not provide better cooperation than that of the simple-standing norm. This suggests that staying can exercise their performance even in private assessment.

3. Discussion

We have developed a systematic method to analyze indirect reciprocity in private assessment without any approximation using solitary observation. So far, only the specified norms can be solved. Previous studies have been unable to solve simple standing, which is one of the leading norms in indirect reciprocity. By assuming that every game is only observed by a single observer,

the equation system can be reduced to two equations (g and g_2, defined in Materials and methods), and thus the number of equations does not continue to infinity. The first equation yields the probability that a discriminating player in the stationary state assesses another player as good. The second equation yields the average probability that two discriminating players (the donor and the observer) simultaneously assess a player as good.

So far, many types of indirect reciprocity in public assessment have an evolutionarily stable point at a state of 100% discriminators (Sasaki et al., 2017). Public assessment guarantees to broadcast a unique assessment to all players, and thus discriminators neither act in a deviant manner nor hurt their images when the system has no errors. On the one hand, unconditional cooperators always cooperate, while discriminators defect if they meet bad players, and thus the discriminators obtain an economic advantage. On the other hand, justified defection by discriminators does not provide any help to all-out defectors. Those two traits lead to a regime consisting of 100% discriminators.

Indirect reciprocity in private assessment provides a different result to that in public assessment. Private assessment does not guarantee that a donor's image of a focal recipient does not necessarily correspond to an observer's image of the recipient. This inconsistency means that the donor's intention of justified defection may not be precisely conveyed to the observer. For example, if both $I_{DR} = B$ and $I_{VR} = G$ are satisfied, the donor, D, did not help the recipient, R. However, from observer, V's point of view, D's action appeared to be malicious because D did not help a good player. This mis-communication means not only that the strict norms including stern judging and shunning have no CES point, but also that even tolerant norms including staying and simple standing cannot maintain a regime with 100% discriminators. This is because defections by discriminators hurt their own images, while unconditional cooperators never hurt their own images, and thus unconditional cooperators can invade a regime of discriminators.

Surprisingly, in simple standing and staying, the average cooperation rate at the CES points in private assessment is greater than that in public assessment. The main reason for this is that unconditional cooperators can raise the cooperation level of the society unless there is no central assessment system or no coercive moral judgment system. Private assessment provides unconditional cooperators with a role.

Our results also show the superiority of the staying norm. Simple standing and staying can maintain cooperative regimes, even in private assessment. These norms incur less information costs than stern judging because those who adopt one of these norms do not need a donor's action if a recipient is assessed as bad. Stern-judging adopters must consider a donor's action if a recipient is bad. Moreover, compared with simple standing, staying has three advantages: it is easier to invade the defectors, it is the more robust to maintain cooperative evolutionarily stable situations, and there is a higher cooperation rate in private assessment. If updating images incur costs, staying has another advantage because staying adopters do not update their private images if a recipient is assessed as bad. An empirical study of staying is an important future extension of this work.

4. Materials and methods

We solve the replicator dynamics of the five norms. First, g is solved in our analysis. Solving g yields each player's payoff (P_X, P_Y, and P_Z), and thus the replicator equations on $\dot{x}$, $\dot{y}$, and $\dot{z}$ are also solved. We then calculate all evolutionarily stable points and solve their conditions.

Table 3
Probability of good in each assessment rule.

Event	Probability
$P_{GB\|C}$	ϵ
$P_{GB\|N}$	e
$P_{GG\|C}$	$1-e$
$P_{GG\|N}$	$1-e$
$P_{BB\|C}$	e
$P_{BB\|N}$	e
$P_{BG\|C}$	$1-\epsilon$
$P_{BG\|N}$	$1-e$

4.1. Influence on assessments of errors

Before analyzing each norm, we calculate the influence on assessments of errors. First, we define $P_{ab|c}$ the probability that an observer assesses a donor as good when the donor's intended action is $c \in \{C, N\}$ and the observer's assessment rule for the donor's cooperation is $a \in \{G, B\}$, while that for the donor's defection is $b \in \{G, B\}$. Table 3 summarizes the calculation in each case. For example, we calculate $P_{GB|C}$. If a donor intends to help a recipient, the intention precisely executes with probability $\bar{e_1}$ and the observer precisely assesses the donor with probability $\bar{e_2}$, while the intention does not execute with probability e_1 and the observer fails the assessment with probability e_2. Therefore, the observer assesses the donor as good with probability ϵ when the donor intends to help the recipient. Therefore, $P_{GB|C} = \epsilon$.

4.2. Image scoring

We calculate g of the image scoring norm (see Table 1). First, we consider g_z. To do so, we consider a game where the donor is a Z player. I_{VD} changes from bad to good if and only if [(1) $I_{VD} = B$ (probability $1 - g_z$)] and [(2) the observer assesses the donor as good] are satisfied. Whether the donor contributes to the recipient depends on I_{DR}. Consider that the recipient is an X player (probability x). If $I_{DR} = G$ (probability g_x), the probability that the observer's assessment is good is $P_{GB|C}$. If $I_{DR} = B$ (probability $1 - g_x$), the assessment is good with probability $P_{GB|D}$. Thus, the probability that I_{VD} changes from bad to good, denoted by g_z^+, is calculated using the following equation. In addition, the probability that I_{VD} changes from good to bad, denoted by g_z^-, is the probability of [Not (1)] and [Not (2)]. Summing up this discussion yields

$$
\begin{aligned}
g_x^+ &= (1-g_x)\epsilon,\\
g_x^- &= g_x(1-\epsilon),\\
g_y^+ &= (1-g_y)e,\\
g_y^- &= g_y(1-e),\\
g_z^+ &= x(1-g_z)[g_x\epsilon + (1-g_x)e]\\
&\quad + y(1-g_z)[g_y\epsilon + (1-g_y)e]\\
&\quad + z(1-g_z)[g_z\epsilon + (1-g_z)e],\\
g_z^- &= xg_z[g_x(1-\epsilon) + (1-g_x)(1-e)]\\
&\quad + yg_z[g_y(1-\epsilon) + (1-g_y)(1-e)]\\
&\quad + zg_z[g_z(1-\epsilon) + (1-g_z)(1-e)].
\end{aligned}
$$

In the steady state, $\Delta g_x (\equiv g_x^+ - g_x^-) = \Delta g_y (\equiv g_y^+ - g_y^-) = \Delta g_z (\equiv g_z^+ - g_z^-) = 0$. These equations yield $g_x = \epsilon$, $g_y = e$ and g_z satisfies $g_z = g(\epsilon - e) + e$ or

$$g_z = \frac{(x\epsilon + ye)(\epsilon - e) + e}{1 - z(\epsilon - e)}$$

.

We then solve the replicator equations. Comparing with Sasaki et al.(2017), the values of g_x, g_y, g_z, and g correspond perfectly to

the case of public information, and thus we skip solving this system.

4.3. Staying

To calculate g for the staying norm (see Table 1), we further define g_{x2}, g_{y2}, and g_{z2} as the fractions of X, Y, and Z players, respectively, that any two Z players assess as good. First, we consider g_z and g_{z2}, and thus we consider a game where the donor is a Z player. I_{VD} changes from bad to good if and only if [(1) $I_{VD}=B$ (probability $1-g_z$)], [(2) $I_{VR}=G$], and [(3) the observer assesses the donor as good]. Consider that the recipient is an X player (probability x). The probability that the observer assesses the donor as good is $P_{GB|C}$ if $I_{DR}=G$ (note that in this case $I_{DR}=I_{VR}=G$, and thus, the probability is g_{x2}). If $I_{DR}=B$ (note that the probability is g_x-g_{x2} because $I_{VR}=G$ and $I_{DR}=B$), the probability is $P_{GB|D}$. Thus, g_z^+ can be calculated using the following equation. In addition, g_z^- is considered if [Not (1)], [(2)] and [Not (3)] are satisfied. We then consider g_{z2}. g_{z2}^+ is the probability that [(1)], [(2)], [(3)], and [(4), i.e. a discriminator who is not chosen in any role in the game assesses the donor as good] are satisfied. Note that the probability of [(1)] and [(4)] is g_z-g_{z2}. In addition, we calculate g_{z2}^-, whose probability is that [Not (1)], [(2)], [Not (3)], and [(4)] are satisfied. Note that the probability of [Not (1)] and [(4)] is g_{z2}. Therefore, the following equations are yielded

$$\begin{aligned}
g_z^+ &= x(1-g_z)[g_{x2}\epsilon+(g_x-g_{x2})e]\\
&\quad +y(1-g_z)[g_{y2}\epsilon+(g_y-g_{y2})e]\\
&\quad +z(1-g_z)[g_{z2}\epsilon+(g_z-g_{z2})e],\\
g_z^- &= xg_z[g_{x2}(1-\epsilon)+(g_x-g_{x2})(1-e)]\\
&\quad +yg_z[g_{y2}(1-\epsilon)+(g_y-g_{y2})(1-e)]\\
&\quad +zg_z[g_{z2}(1-\epsilon)+(g_z-g_{z2})(1-e)],\\
g_{z2}^+ &= x(g_z-g_{z2})[g_{x2}\epsilon+(g_x-g_{x2})e]\\
&\quad +y(g_z-g_{z2})[g_{y2}\epsilon+(g_y-g_{y2})e]\\
&\quad +z(g_z-g_{z2})[g_{z2}\epsilon+(g_z-g_{z2})e],\\
g_{z2}^- &= xg_{z2}[g_{x2}(1-\epsilon)+(g_x-g_{x2})(1-e)]\\
&\quad +yg_{z2}[g_{y2}(1-\epsilon)+(g_y-g_{y2})(1-e)]\\
&\quad +zg_{z2}[g_{z2}(1-\epsilon)+(g_z-g_{z2})(1-e)].
\end{aligned}$$

Calculating g_x, g_{x2}, g_y, and g_{y2} is easier than in the case of Z because [(3)] is constant irrespective of the type of recipient, thus

$$\begin{aligned}
g_x^+ &= (1-g_x)g\epsilon,\\
g_x^- &= g_xg(1-\epsilon),\\
g_{x2}^+ &= (g_x-g_{x2})g\epsilon,\\
g_{x2}^- &= g_{x2}g(1-\epsilon),\\
g_y^+ &= (1-g_y)ge,\\
g_y^- &= g_yg(1-e),\\
g_{y2}^+ &= (g_y-g_{y2})ge,\\
g_{y2}^- &= g_{y2}g(1-e).
\end{aligned}$$

In the steady state, $\Delta g_x=\Delta g_{x2}(\equiv g_{x2}^+-g_{x2}^-)=\Delta g_y=\Delta g_{y2}(\equiv g_{y2}^+-g_{y2}^-)=\Delta g_z=\Delta g_{z2}(\equiv g_{z2}^+-g_{z2}^-)=0$. These equations yield $g_x=\epsilon$, $g_y=e$, $g_{x2}=\epsilon^2$, $g_{y2}=e^2$, and $g_{z2}=g_z^2$ and g_z satisfies

$$(g_z-e)g=(\epsilon-e)g_2, \tag{1}$$

where $g_2=xg_x^2+yg_y^2+zg_z^2$.

We then solve the replicator system and derive the following theorem.

Theorem 1. *In staying, if the following condition (Eq. (2)) is satisfied, the system has two locally asymptotically stable fixed points at* $(x,y,z)=(0,1,0)$ *and* $Q(1-z^*,0,z^*)$, *where* $P_Z=P_X$ *is satisfied at point* Q.

$$r>\frac{1}{\epsilon-e} \tag{2}$$

Proof. First, as shown in Lemma 1, there is no inner fixed point. This also yields no closed orbit (and thus any limit cycle) because any closed orbit in a two-dimensional continuous mapping absolutely requires a fixed point inside the orbit. It is trivial that (0,1,0) is a locally asymptotically stable fixed point because both $P_Y>P_X$ and $P_Y>P_Z$ are absolutely satisfied at the point and its neighborhood. Second, the dynamics on the edges are calculated. It is trivial that $P_Y>P_X$ on the line of $z=0$. This feature and Lemma 2 show that it is enough to consider the line of $y=0$. Finally, that a point of $z=z^*$ is an attractor on the line of $y=0$ is shown using Lemma 3. End of proof. □

Lemma 1. *The system has no inner fixed point.*

Proof. Suppose that (x^*, y^*, z^*) is an inner fixed point and g_x^*, g_y^*, g_z^*, g^*, and g_2^* are the values of g_x, g_y, g_z, g, and g_2, respectively, at the point. $P_X=P_Y$ yields $rz^*(\epsilon-e)=1$ and $P_Y=P_Z$ yields $rz^*(g_z^*-e)=g^*$. Substituting these equations into Eq. (1), $g_2^*=(g^*)^2$. However, $g_2^*-(g^*)^2=x^*y^*(g_x^*-g_y^*)^2+y^*z^*(g_y^*-g_z^*)^2+z^*x^*(g_z^*-g_x^*)^2>0$ is always satisfied because $x^*y^*z^*\neq 0$. Contradiction. End of proof. □

Lemma 2. *The dynamics of the line of* $x=0$ *is bi-stable.*

Proof. When $x=0$, $g=(1-z)e+zg_z$ and $g_2=(1-z)e^2+zg_z^2$ are satisfied. Substituting these equations, Eq. (1) is replaced by $(1-\epsilon+e)zg_z^2+e(1-2z)g_z-e^2(1-\epsilon+e)(1-z)=0$, and thus, if $z\neq 0$ then

$$g_z=\frac{e\sqrt{D(z)}-e(1-2z)}{2(1-\epsilon+e)z}, \tag{3}$$

and if $z=0$ then $g_z=e(1-\epsilon+e)$ where $D(z)=1-4z(1-z)K_1$ and $K_1=1-(1-\epsilon+e)^2$. Let $f(z)\equiv P_Z-P_Y$. This yields

$$f(z)=(r-1)z(g_z-e)-e. \tag{4}$$

Substituting Eq. (3) into Eq. (4) yields

$$f(z)=\frac{(r-1)e}{2(1-\epsilon+e)}[\sqrt{D(z)}+2z(\epsilon-e)-1]-e.$$

If $f=0$, then $\sqrt{D(z)}=K_2-2z(\epsilon-e)$, where $K_2=\frac{2(1-\epsilon+e)}{r-1}+1$, and thus $(f_1(z)\equiv)4[K_1-(\epsilon-e)^2]z^2-4[K_1-K_2(\epsilon-e)]z+1-K_2^2=0$ is satisfied. This quadratic function has a unique solution in $0<z<1$ because $f_1(0)<0$ and $f_1(1)=\frac{4r(1-\epsilon+e)}{(r-1)^2}[r(\epsilon-e)-1]>0$ if $r(\epsilon-e)>1$. f is continuous and $f(0)<0$ and $f(1)>0$ are satisfied if $r(\epsilon-e)>1$. End of proof. □

Lemma 3. $Q(1-z^*,0,z^*)$ *is a local attractor.*

Proof. Suppose that $y=0$. In addition to Lemma 2, we will first prove that a continuous function, $f(z)\equiv P_Z-P_X=0$, has a unique solution, $z=z^*$, in $0<z<1$ and $f(0)>0$ and $f(1)<0$ are satisfied when $r(\epsilon-e)>1$. Let g_x^*, g_y^*, g_z^*, g^*, and g_2^* be the values of g_x, g_y, g_z, g, and g_2, respectively, when $z=z^*$. Substituting $g^*=(1-z^*)\epsilon+z^*g_z^*$ into $f(z^*)=0$ yields

$$z^*(\epsilon-g_z^*)=\frac{1-\epsilon}{r-1}. \tag{5}$$

Substituting Eq. (5) and $g_2^*=(1-z^*)\epsilon^2+z^*(g_z^*)^2$ into Eq. (1) yields

$$g_z^*=\frac{(r\epsilon-1)[e+\epsilon(\epsilon-e)]}{r\epsilon-1+(1-\epsilon)(\epsilon-e)}. \tag{6}$$

Therefore,

$$z^* = \frac{r\epsilon - 1 + (1-\epsilon)(\epsilon - e)}{(r-1)(\epsilon - e)(r\epsilon - 1 + \epsilon)}. \tag{7}$$

If $r(\epsilon - e) > 1$ is satisfied, $0 < z^* < 1$ is also satisfied.

Second, we will prove that Y cannot invade point Q. Substituting Eq. (7),

$$P_X(Q) - P_Y(Q) = \frac{(1-\epsilon)[r(\epsilon - e) - 1]}{(r-1)(r\epsilon - 1 + \epsilon)} > 0$$

if $r(\epsilon - e) > 1$ is satisfied. End of proof. □

4.4. Simple-standing

Consider g_z and g_{z2}, and thus the case where the donor is a Z player. I_{VD} changes from bad to good if and only if [(1) $I_{VD} = B$ (probability $1 - g_z$)], (([(2) $I_{VR} = B$], and [(3) no assessment error occurs]) or ([(4) $I_{VR} = G$] and [(5) the observer assesses the donor as good])). The probability that the observer's assessment of the donor is good is $P_{GB|C}$ if $I_{DR} = G$ (note that the probability of the case $I_{VR} = I_{DR} = G$ is g_2). If $I_{DR} = B$ (note that the probability is $g - g_2$), the probability is $P_{GB|D}$. In addition, I_{VD} changes from bad to bad if and only if [Not (1)] and (([2] and [Not (3)]) or ([(4)] and [Not (5)])). We then consider g_{z2}. g^+_{z2} is the probability of [(1)], (([(2)] and [(3)]) or ([(4)] and [5])), and ([6] that a discriminator who is not chosen in any role in the game assesses the donor as good) are satisfied. Note that the probability of [(1)] and [(6)] is $g_z - g_{z2}$. g^-_{z2} is the probability of [Not (1)] and (([2] and [Not (3)]) or ([(4)], [Not (5)])), and [(6)]. Note that the probability of [Not (1)] and [(6)] is g_{z2}. Therefore,

$$\begin{aligned}
g^+_z &= (1-g_z)[(1-g)(1-e) + g_2\epsilon + (g-g_2)e],\\
g^-_z &= g_z[(1-g)e + g_2(1-\epsilon) + (g-g_2)(1-e)],\\
g^+_{z2} &= (g_z - g_{z2})[(1-g)(1-e) + g_2\epsilon + (g-g_2)e],\\
g^-_{z2} &= g_{z2}[(1-g)e + g_2(1-\epsilon) + (g-g_2)(1-e)].
\end{aligned}$$

Calculating g_x, g_{x2}, g_y and g_{y2} is easier than in the case of Z because [(5)] is constant irrespective of the type of recipient. Thus,

$$\begin{aligned}
g^+_x &= (1-g_x)[(1-g)(1-e) + g\epsilon],\\
g^-_x &= g_x[(1-g)e + g(1-\epsilon)],\\
g^+_{x2} &= (g_x - g_{x2})[(1-g)(1-e) + g\epsilon],\\
g^-_{x2} &= g_{x2}[(1-g)e + g(1-\epsilon)],\\
g^+_y &= (1-g_y)[(1-g)(1-e) + ge],\\
g^-_y &= g_y[(1-g)e + g(1-e)],\\
g^+_{y2} &= (g_y - g_{y2})[(1-g)(1-e) + ge],\\
g^-_{y2} &= g_{y2}[(1-g)e + g(1-e)].
\end{aligned}$$

In the steady state, these equations yield $g_{x2} = g_x^2$, $g_{y2} = g_y^2$, $g_{z2} = g_z^2$, and

$$\begin{aligned}
g_x &= (1-g)(1-e) + g\epsilon,\\
g_y &= (1-g)(1-e) + ge,\\
g_z &= (1-g)(1-e) + g_2\epsilon + (g-g_2)e
\end{aligned} \tag{8}$$

As well as staying, we solve the replicator system and derive the following theorem.

Theorem 2. *In simple-standing, if the following condition (Eq. (9)) is satisfied, the system has two locally asymptotically stable fixed points at $(x, y, z) = (0, 1, 0)$ and $Q(1 - z^*, 0, z^*)$, where $P_X = P_Z$ is satisfied at point Q.*

$$r > \frac{1}{\epsilon - e}\left(1 + \sqrt{\frac{1-\epsilon}{1-e}}\right) \tag{9}$$

Proof. We prove this theorem using the following lemmas (Lemmas 4–6, as well as the same approach used in the proof of Theorem 2. The following facts are useful for the proof. $(x, y, z) = (0, 1, 0)$ is locally stable because $g = g_y$ yields $g = 1/2$ and $(P_X, P_Y, P_Z) = (-1, 0, -1/2)$. When $z = 0$, $P_X = rx - 1 < P_Y = rx$. End of proof. □

Lemma 4. *The system has no inner fixed point.*

Proof. Suppose that (x^*, y^*, z^*) is an inner fixed point. $P_X = P_Y$ yields $rz^*(g^*_x - g^*_y) = 1$ and $P_Y = P_Z$ yields $rz^*(g^*_z - g^*_y) = g^*$. Substituting these equations into Eq. (8), $g^*_2 = (g^*)^2$. In addition to Lemma 1, $g^*_2 > (g^*)^2$ is satisfied because of $x^*y^*z^* \neq 0$. Contradiction. End of proof. □

Lemma 5. *The dynamics of the line of $x = 0$ is bi-stable.*

Proof. Suppose $x = 0$. The definitions of g and g_2, respectively, are replaced by

$$g = (1-z)g_y + zg_z \tag{10}$$

and

$$g_2 = (1-z)g_y^2 + zg_z^2. \tag{11}$$

Substituting Eq. (10) into Eq. (8) yields

$$g = (1-g)(1-e) + (1-z)ge + z[g_2\epsilon + e(g-g_2)]. \tag{12}$$

Eq. (12) yields

$$g_2 z(\epsilon - e) = (2\,g - 1)(1-e), \tag{13}$$

and thus, $g \geq 1/2$ must be satisfied.

Substituting Eq. (8) into Eq. (10) yields

$$zg_z = (1-e)(1-z)(2g-1) + zg. \tag{14}$$

Substituting Eqs. (8) and (14) into Eq. (11) yields

$$zg_2 = (1-e)^2(1-z)(2g-1)^2 + zg^2. \tag{15}$$

Comparing Eq. (13) with Eq. (15),

$$\begin{aligned}
f_1(z) \equiv\ & (\epsilon - e)(1-e)^2(1-z)(2g-1)^2\\
& + (\epsilon - e)zg^2 - (2g-1)(1-e) = 0
\end{aligned} \tag{16}$$

is satisfied, and thus, $g = g(z)$. $\frac{d}{dz}f_1 = 0$ yields $L_1 g' = L_2$ where

$$L_1 = 4(\epsilon - e)(1-e)^2(1-z)(2g-1) + 2(\epsilon - e)zg - 2(1-e),$$

and

$$L_2 = -(\epsilon - e)[(1-e)(2g-1) + g][1 - g + e(2g-1)].$$

$L_2 < 0$ is trivial. $L_1 < 0$ when $g > 1/2$ because substituting Eqs. (16) and (13) into $L_1(2g-1)$ yields

$$L_1(2g-1) = 2[(1-e)(2g-1) - (\epsilon - e)zg] = 2(\epsilon - e)z(g_2 - g).$$

Therefore, $g = g(z)$ is a monotonically increasing function on z.

Let $f(z) \equiv P_Z - P_Y$. Using Eqs. (8) and (13), f is replaced by

$$f(z) = g[2r(1-e) - 1] - r(1-e),$$

and thus $f = 0$ has a unique solution. Using $g(z = 0) = 1/2$, $f(z = 0) = -1/2 < 0$. Moreover, if Eq. (9) is satisfied, using $g(z = 1) = \frac{1-e-\sqrt{(1-e)(1-\epsilon)}}{\epsilon - e}$,

$$f(z = 1) = \frac{(1-e)[r(2-\epsilon-e) - 1] - \sqrt{(1-e)(1-\epsilon)}[2r(1-e) - 1]}{\epsilon - e} > 0.$$

End of proof. □

Lemma 6. *$Q(1 - z^*, 0, z^*)$ is a local attractor.*

Proof. Suppose $y = 0$. In addition to Lemma 5, g_2 can be cleaned by solving Eq. (8) and the definitions of g and g_2. Substituting Eq. (8) into $g = (1-z)g_x + zg_z$ yields

$$zg_2(\epsilon - e) = g - (1-z)g_x - z(1-g-e). \tag{17}$$

Using $g_2 = (1-z)g_x^2 + zg_z^2$,

$$zg_2 = (1-z)g_x - 2 + g^2 - 2(1-z)zgg_x. \tag{18}$$

Comparing Eq. (17) with Eq. (18),

$$z = \frac{(g_x - g)^2 + \frac{g_x - g}{\epsilon - e}}{(g_x - g)^2 + g(1-g)}. \tag{19}$$

Let $f(z)$ be $P_Z - P_X$. Then using Eq. (8) and the definition of g,

$$f(z) = 1 - g + r(1-\epsilon)g - r(1-g)(1-e).$$

When $f = 0$, g can be solved as a unique constant in $0 < g(\equiv g^*) < 1$ where

$$g^* = \frac{r(1-e) - 1}{r(2-\epsilon-e) - 1}.$$

Using Eq. (19), there is a unique solution $z = z^*$ when $g = g^*$, where

$$z^* = \frac{1 - \epsilon + \frac{r(2-\epsilon-e)-1}{\epsilon - e}}{1 - \epsilon + r^2(1-e) - r}.$$

$0 < z^* < 1 \leftrightarrow b^2(\epsilon - e)(1-e) - 2bc(1-e) + c^2 > 0$. Note that $1 > \frac{1-\sqrt{\frac{1-\epsilon}{1-e}}}{\epsilon - e}$, and thus Eq. (9) must be satisfied.

Using $g(z=0) = \frac{1-e}{2-e-\epsilon}$, $f(0) = \frac{1-\epsilon}{2-e-\epsilon} > 0$, and using $g(z=1) = \frac{1-e-\sqrt{(1-e)(1-\epsilon)}}{\epsilon - e}$, $f(1) < 0$ if and only if Eq. (9) is satisfied. The function f is a continuum, and thus it is proved that Q is an attractor if $y = 0$.

Second, we will prove that $P_X > P_Y$ at point Q. Using the values of z^* and g^*,

$$P_X(Q) - P_Y(Q) = \frac{r^2(\epsilon - e)(1-e) - 2r(1-e) + 1}{[r(2-\epsilon-e) - 1][r^2(1-e) - r + 1 - \epsilon]}(1-\epsilon).$$

This is positive if Eq. (9) is satisfied. End of proof. □

4.5. Stern judging

In addition to the previous norms, the system yields $g_{x2} = g_x^2$, $g_{y2} = g_y^2$, $g_{z2} = g_z^2$, and

$$g_x = (1-2g)(1-\epsilon) + g,$$
$$g_y = (1-2g)(1-e) + g,$$
$$g_z = (1-2g)(1-e) + g + (\epsilon - e)(2g_2 - g)$$

in the steady state. As well as the above norms, we solve the replicator system and derive the following Theorem 3.

Theorem 3. *In stern judging, $g = g_x = g_y = g_z = 1/2$ is always satisfied regardless of the values of (x, y, z), and thus $P_Y > P_Z > P_X$ is always satisfied. Therefore, the system has a globally asymptotically stable fixed point at $(x, y, z) = (0, 1, 0)$.*

Proof. On the one hand, the equation for g_2 is yielded using the definition of g_z and $g = xg_x + yg_y + zg_z$ as

$$2zg_2 = zg + (2g-1)(p-x)$$

where $p = \frac{1-e}{\epsilon - e}$. On the other hand, another equation for g_2 is also yielded using the following definition of g_2:

$$\begin{aligned} zg_2 &= xzg_x^2 + yzg_y^2 + (g - xg_x - yg_y)^2 \\ &= zg^2 + (1-2g)^2 L, \end{aligned}$$

where $L = x(1-\epsilon)^2 + y(1-e)^2 - xy(\epsilon - e)^2$. Note that $L \geq 0$ because L is an increasing linear function on y for a given x, and thus, $L \geq L(y=0) = x(1-\epsilon)^2 \geq 0$. Using these two equations on g_2, $g = \frac{1}{2}$, $\frac{p-x+2L}{z+2L}(\equiv g_*)$. If $(f \equiv)p - x + 2L - (z+4L) > 0$ is satisfied, then $g_* > 1$, and thus, g_* is beyond the scope of the definition. f is also an increasing linear function on y for a given x, and thus, $f \geq f(y=0) = p[1 - 2x(\epsilon - e)(1-\epsilon)] > 0$. End of proof. □

4.6. Shunning

In addition to the previous norms, the system yields $g_{x2} = g_x^2$, $g_{y2} = g_y^2$, $g_{z2} = g_z^2$, and

$$\begin{aligned} g_x &= g(\epsilon - e) + e, \\ g_y &= e, \\ g_z &= g_2(\epsilon - e) + e \end{aligned} \tag{20}$$

in the steady state. In addition to the above norms, we solve the replicator system and derive the following theorem.

Theorem 4. *In shunning, if the following condition (Eq. (21)) is satisfied, the system has two locally asymptotically stable fixed points at $(x, y, z) = (0, 1, 0)$ and (0,0,1), otherwise the system has a globally asymptotically stable fixed point at (0,1,0).*

$$r > \frac{1 + \sqrt{1 - 4e(\epsilon - e)}}{2e(\epsilon - e)} \tag{21}$$

Proof. We prove this using the following lemmas (Lemmas 7–9) and $P_X + c = P_Y$ when $z = 0$. End of proof. □

Lemma 7. *The system has no inner fixed point.*

Proof. Proof. Suppose that (x^*, y^*, z^*) is an inner fixed point. $P_X = P_Y$ yields $rz^*(g_x^* - g_y^*) = 1$ and $P_Y = P_Z$ yields $rz^*(g_z^* - g_y^*) = g^*$. Substituting these equations into Eq. (20), $g_2^* = (g^*)^2$. Contradiction. End of proof. □

Lemma 8. *The dynamics of the line of $x = 0$ is bi-stable.*

Proof. Proof. Suppose that $x = 0$. When $z = 0$, $g = g_y$=e is satisfied, and thus, $P_Z - P_Y = -ce < 0$ is satisfied. Hence, suppose that $z > 0$. The definitions of g of g_2, respectively, are replaced by

$$g = (1-z)g_y + zg_z = e + zg_2(\epsilon - e) \tag{22}$$

and

$$g_2 = (1-z)e^2 + z[g_2(\epsilon - e) + e]^2. \tag{23}$$

Eq. (23) is a quadratic function on g_2, and thus g_2 is solved as

$$g_2 = \frac{1 - 2ze(\epsilon - e) - \sqrt{D(z)}}{2z(\epsilon - e)^2},$$

where $D(z) = 1 - 4ze(\epsilon - e) - 4z(1-z)e^2(\epsilon - e)^2$. Substituting this solution into Eq. (22) yields $g = e + \frac{1 - f(z)}{2(\epsilon - e)}$ where $f(z) = 2ze(\epsilon - e) + \sqrt{D(z)}$. f is a decreasing function on z because

$$f'(z) = \frac{2e(\epsilon - e)}{\sqrt{D(z)}}[\sqrt{D(z)} + e(\epsilon - e)(2z - e) - 1] < 0.$$

This means that g is an increasing function on z.

Using Eq. (22),

$$P_Z - P_Y = rz(g_z - g_y) - g = rzg_2(\epsilon - e) - g = r(g - e) - g,$$

and thus $P_Z - P_Y$ is also an increasing function on z. When $z = 1$,

$$P_Z(z=1) - P_Y(z=1) = \frac{1 - \sqrt{1 - 4e(\epsilon - e)}}{2(\epsilon - e)}(r-1) - re > 0$$

if the condition in Eq. (21) is satisfied. End of proof. □

Lemma 9. *If $y = 0$ then $P_X < P_Z$.*

Proof. $g \geq g_2$ is satisfied, and thus, the definition of g is replaced by

$$\begin{aligned} g &= (1-z)g_x + zg_z \\ &= (1-z)[g(\epsilon-e)+e] + z[g_2(\epsilon-e)+e] \\ &\le (1-z)[g(\epsilon-e)+e] + z[g(\epsilon-e)+e] \\ &= g(\epsilon-e)+e, \end{aligned} \tag{24}$$

and thus

$$g \le \frac{e}{1-\epsilon+e} \tag{25}$$

is always satisfied.

Substituting Eqs. (24) and (25),

$$\begin{aligned} P_X - P_Z &= rz(g-g_2)(\epsilon-e) - 1 + g \\ &= re - rg(1-\epsilon+e) - 1 + g \\ &= g[1-r(1-\epsilon+e)] + re - 1 \\ &\le \frac{e}{1-\epsilon+e}[1-r(1-\epsilon+e)] + re - 1 \\ &= \frac{-1+\epsilon}{1-\epsilon+e} < 0. \end{aligned}$$

End of proof. □

Competing interests

The authors declare no competing interests. T.S. is an employee of F-power Inc. The other authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Acknowledgments

We thank Christian Hilbe for his comments. This work was supported by JSPS KAKENHI grant numbers 16H03120 and 17H02044 (to I.O), Austrian Science Fund (FWF) Grant P27018-G11 (to T.S.), and JSPS KAKENHI grant numbers 16H03698 (to Y.N.).

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